Camera–LiDAR Fusion for Object Detection on KITTI

3D Perception & Learning-Based Data Fusion · Project 2

Abstract

Building on the calibration and 3D-box geometry of Project 1, this project predicts objects from raw sensor data and scores them against the KITTI annotations. Three classical detectors (camera-only, LiDAR-only and fusion) for Car, Van, Truck, Pedestrian and Cyclist reach a mAP of 2.2%, 60.8% and 62.0% on 600 unseen frames.

1. Data and ground truth

The 7,481 labelled KITTI training frames [1] are split into disjoint groups: 3,000 to train the camera models, 2,900 to train the LiDAR and fusion classifiers, and 600 for evaluation only. Camera scores fed to the fusion classifier are produced out-of-fold, so no model is scored on data it has seen.

Ground truth is partial. Only objects in the left camera's view are labelled, distant or tiny ones are often unlabelled or marked DontCare, and boxes contain annotator noise. A detection matching no label may be a real unlabelled object, so precision is measured *against the labels*. Without any labels, AP and recall could not be computed; we would hand-label a small subset and report metrics on it with their uncertainty, plus clearly flagged proxy checks (sensor agreement, temporal consistency).

2. Methods

Camera-only: HOG features [2] and one linear SVM per class, slid over an eight-level image pyramid, with three rounds of hard-negative mining, box regression and non-maximum suppression. **LiDAR-only:** RANSAC [4] removes the road, DBSCAN [3] clusters the remaining points into candidates, and a random forest [5] classifies each cluster from about 30 geometric features (size, height profile, distance, density, reflectance); each cluster's projected 2D box is refined by a learned regressor. **Fusion:** the same LiDAR candidates, classified by a forest that also receives camera evidence for each box (SVM scores and an image-crop classifier trained on cluster boxes). Fusion is therefore feature-level and LiDAR-led.

3. Evaluation protocol

A detection is correct if its 2D IoU with an unmatched true box of the same class is ≥ 0.5. We report precision, recall, AP (40-point KITTI interpolation) and mAP under KITTI *moderate* difficulty. **Deviation:** the official benchmark uses IoU 0.7 for cars and 3D boxes; we use 2D IoU 0.5 for all classes because the camera detector has no 3D output. To locate failures, we report *cluster recall* (share of true objects with any LiDAR candidate box, the ceiling for LiDAR-led detection) and rescore at IoU ≥ 0.3 to separate "not found" from "loosely boxed".

4. Results

Table 1. AP (%) on 600 validation frames, IoU ≥ 0.5, moderate difficulty.

Detector	Car	Van	Truck	Pedestrian	Cyclist	mAP
Camera-only	8.9	0.8	0.4	0.4	0.7	2.2
LiDAR-only	87.1	56.2	42.0	45.9	72.9	60.8
Fusion	88.1	59.2	42.6	46.5	73.5	62.0

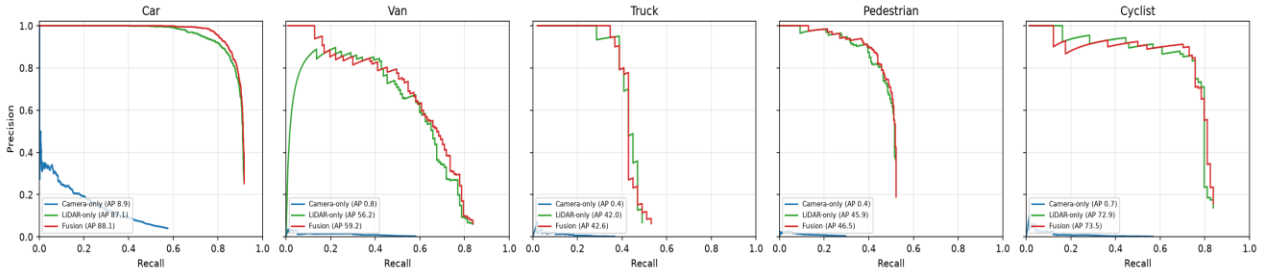


Figure 1. Precision–recall curves: camera-only (blue), LiDAR-only (green), fusion (red).

Table 2. Cluster recall: share of valid true objects with a LiDAR candidate box.

	Car	Van	Truck	Pedestrian	Cyclist
True objects	1278	117	49	293	74
IoU ≥ 0.5	92.3%	87.2%	59.2%	54.6%	85.1%
IoU ≥ 0.3	95.9%	94.0%	63.3%	78.2%	89.2%



Figure 2. Frame 004973 (ground truth in green): the camera fires on clutter and misses the pedestrian; fusion finds it.

5. Critical analysis

Camera. One HOG template per class must cover every viewpoint, objects are often 25–40 px tall, and car, van and truck look alike; maximum car recall is only 58%.

LiDAR. Geometry isolates objects directly, so cars work well (cluster recall 92%). Pedestrians fail on box quality: a cluster exists for 78% at IoU ≥ 0.3 but only 55% at 0.5. Trucks fail on missing clusters (63% even at 0.3), when partly out of view or occluded. No classifier can recover an object that was never proposed, so fusion shares this ceiling.

Fusion. The gain is small and comes through precision: best-F1 precision for cars rises from 0.89 to 0.92 at equal recall (Van AP +3.0), as the camera rejects clutter and separates vans from cars; camera evidence carries 26% of the forest's feature importance. Box refinement mattered most, raising LiDAR mAP from 48.1% to 60.8%.

Limitations. With only 117 vans, 74 cyclists and 49 trucks, one-point AP differences are within noise; the validation frames also guided development, and 3D accuracy is not yet evaluated.

6. Conclusion

LiDAR dominates in this classical setting, and camera evidence adds a small precision gain. These results are the baseline for Project 3, which replaces hand-designed features with learning-based fusion using neural networks, so that one network learns from both sensors jointly. Project 3 will also predict full 3D bounding boxes (position, size and orientation) for each object, extending the oriented cuboids of Project 1 from labels to model predictions, evaluated against the 3D annotations, with deep detectors such as PointPillars [6] as the reference.

References

[1] A. Geiger et al., "Are we ready for autonomous driving? The KITTI vision benchmark suite," CVPR, 2012. [2] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," CVPR, 2005. [3] M. Ester et al., "A density-based algorithm for discovering clusters…," KDD, 1996. [4] M. A. Fischler and R. C. Bolles, "Random sample consensus," Comm. ACM, 1981. [5] L. Breiman, "Random forests," Machine Learning, 2001. [6] A. H. Lang et al., "PointPillars," CVPR, 2019.